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April 21, 2009

Recommendation Letter

This is to recommend Igor Netai for participation in Summer School "*Structures in Lie Representation Theory*".

I know Igor since September 2005. He was among the best students in my algebra classes for the first and later for the second year students at Moscow State University.

Since September 2007, Igor is a student of Higher Algebra Department, and I am his scientific supervisor. Current Igor's work is devoted to the theory of affine algebraic groups. Let G be a connected semisimple group. A subgroup $H \subseteq G$ is called *parabolically connected* if the intersection of H with any parabolic subgroup $P \subseteq G$ is connected. The motivation for study of this class of subgroups comes from the paper (J.Hausen, Algebraicity criteria for almost homogeneous complex spaces, Arch. Math 74 (2000), 317-320), where in similar terms a sufficient condition for a quasihomogeneous Moishezon space to be an algebraic variety is obtained. Igor considers the case when H is a connected reductive spherical subgroup of a simple group. Using Kraemer's classification he has proved that any such subgroup in $\mathrm{SL}(n)$ except for $\mathrm{SO}(n)$ is parabolically connected. This is a non-trivial result, where some elegant and new lemmas from linear algebra are needed. The text is in preparation now.

In my opinion, Igor Netai is a gifted young mathematician. I think that he will fully benefit from attending the summer school, and I strongly recommend Igor for this event.